

1. INTRODUCTION

1.1 Purpose

This document provides a comprehensive description of the **RFP Document Processor** application. It defines the functional and non-functional requirements for the AI-powered tool that analyzes RFP documents against a company checklist and provides Go/No-Go recommendations.

1.2 Scope

The RFP Document Processor is a web-based application that:

- Accepts RFP documents in multiple formats (PDF, DOCX, TXT)
- Extracts text from uploaded documents
- Uses Google Gemini AI to analyze content against a company checklist
- Evaluates 19 criteria across Financial, Legal, Operations, and Technical departments
- Provides a Go/No-Go decision with a percentage score
- Displays detailed checklist results in a tabular format
- Exports results as JSON

1.3 Definitions and Acronyms

Term	Definition
RFP	Request for Proposal
GO	Decision to bid on the RFP
NO-GO	Decision not to bid
ESCALATE	Decision requiring management review
PDF	Portable Document Format

Term	Definition
DOCX	Microsoft Word Document
TXT	Plain text file

2. OVERALL DESCRIPTION

2.1 Product Perspective

The RFP Document Processor is a standalone web application built using Streamlit. It integrates with Google Gemini AI for document analysis. The application is deployed on Streamlit Cloud, accessible from any web browser.

2.2 User Characteristics

User Role	Description
Proposal Manager	Reviews RFPs and makes bid decisions
Legal Team	Reviews compliance and contract terms
Accounting Team	Reviews financial requirements
Technical Team	Reviews technical specifications

2.3 Operating Environment

- **Platform:** Web-based application
 - **Framework:** Streamlit (Python)
 - **AI Service:** Google Gemini API
 - **File Storage:** Temporary storage only (no database)
 - **Hosting:** Streamlit Cloud or local deployment
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3. FUNCTIONAL REQUIREMENTS

3.1 Document Upload

ID	Requirement	Priority
FR-01	Accept file uploads in PDF, DOCX, and TXT formats	High
FR-02	Extract text from PDF using PyPDF2	High
FR-03	Extract text from DOCX using python-docx	High
FR-04	Clean extracted text (remove page numbers, fix spacing)	Medium
FR-05	Display file name and size after upload	Low

3.2 AI Analysis

ID	Requirement	Priority
FR-06	Integrate with Google Gemini API	High
FR-07	Extract project summary from RFP	High
FR-08	Extract deliverables list	High
FR-09	Extract evaluation criteria	High
FR-10	Extract compliance checklist by department	High
FR-11	Evaluate 19 checklist criteria (0-10 each)	High
FR-12	Calculate overall percentage score (0-100)	High

ID	Requirement	Priority
FR-13	Determine GO/NO-GO/ESCALATE based on score	High

3.3 Checklist Criteria

Financial Department:

1. Payment Terms - NET30 or better
2. Insurance Requirements - \$5M or less
3. Financial Stability - Audited statements
4. Profitability - Budget known
5. Bid Bond - Can provide

Legal Department:

6. Eligibility Criteria - Meet requirements
7. State Registration - Have registration
8. E-Verify - Have verification
9. Contract Terms - Acceptable terms
10. Legal Compliance - Comply with laws

Operations Department:

11. Required Forms - Standard forms
12. Submission Deadlines - Feasible timeline
13. Signatory Authority - Available
14. Vendor Registration - Have registration

Technical Department:

15. Scope Alignment - Match services
16. Technical Requirements - Meet specifications
17. Industry Standards - Comply with standards
18. Security Requirements - Meet security needs

19. Integration Needs - Can integrate

3.4 Decision Logic

Score Range	Decision	Meaning
71-100	GO	We should bid
51-70	ESCALATE	Management review needed
0-50	NO-GO	We should not bid

3.5 Results Display

ID	Requirement	Priority
FR-14	Display big decision box with color coding	High
FR-15	Show overall score as rounded integer	High
FR-16	Display progress bar for score visualization	High
FR-17	Show metrics for GO/NO-GO/ESCALATE counts	High
FR-18	Display checklist results in departmental tables	High
FR-19	Show project summary, deliverables, evaluation criteria	Medium

3.6 Multi-RFP Comparison

ID	Requirement	Priority
FR-20	Support uploading 2-5 documents simultaneously	High

ID	Requirement	Priority
FR-21	Process all documents with progress indicator	High
FR-22	Display comparison table with all RFPs	High
FR-23	Highlight Best RFP with dark green color	High
FR-24	Highlight Worst RFP with dark red color	High
FR-25	Provide tabs for detailed results per RFP	High
FR-26	Export all results as JSON	Medium

3.7 Export

ID	Requirement	Priority
FR-27	Download results as JSON	Medium
FR-28	Include checklist items, decision, score in export	Medium

3.8 Configuration

ID	Requirement	Priority
FR-29	Accept Gemini API key via sidebar	High
FR-30	Store API key using Streamlit Secrets	High
FR-31	Display API key status (provided/missing)	Medium

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance

ID	Requirement	Priority
NFR-01	Document processing within 60 seconds	High
NFR-02	Handle files up to 50 MB	High
NFR-03	UI response within 2 seconds	High

4.2 Reliability

ID	Requirement	Priority
NFR-04	Gracefully handle API quota errors (429)	High
NFR-05	Provide fallback responses when AI fails	High
NFR-06	Clean up temporary files after processing	High

4.3 Security

ID	Requirement	Priority
NFR-07	API keys not exposed in code or logs	High
NFR-08	Uploaded files deleted after processing	High
NFR-09	No user data stored permanently	High

4.4 Usability

ID	Requirement	Priority
NFR-10	Intuitive UI with clear instructions	High
NFR-11	Results with clear color coding	High
NFR-12	User-friendly error messages	High

5. COLOR CODING

Decision	Background	Border	Text
GO	Dark Green (#1e7e34)	Green (#28a745)	White
NO-GO	Dark Red (#721c24)	Red (#dc3545)	White
ESCALATE	Dark Yellow (#856404)	Yellow (#ffc107)	White

6. TECHNOLOGY STACK

Component	Technology	Version
Frontend/Backend	Python	3.14
Web Framework	Streamlit	1.28.0
AI Service	Google Gemini	gemini-1.5-flash
PDF Processing	PyPDF2	3.0.1

Component	Technology	Version
DOCX Processing	python-docx	1.1.0

7. ERROR HANDLING

Scenario	Response
No API key	"Please provide your Gemini API key in the sidebar"
API quota exceeded	"Quota exceeded. Please wait or try a different API key"
Unsupported file	"Unsupported format. Upload PDF, DOCX, or TXT"
Empty/corrupted file	"No text could be extracted from the document"
AI fails	"AI encountered an error. Please review the RFP manually"

8. ACCEPTANCE CRITERIA

Criterion	Description
AC-01	Deploys successfully on Streamlit Cloud
AC-02	Accepts PDF, DOCX, and TXT uploads
AC-03	Extracts text from uploaded documents
AC-04	Processes documents using Gemini API

Criterion	Description
AC-05	Displays GO/NO-GO with correct color coding
AC-06	Calculates score correctly (0-100)
AC-07	Shows checklist table with evidence
AC-08	Supports multi-RFP comparison
AC-09	Exports results as JSON

9. DEPENDENCIES

- Google Gemini API key
 - Internet connection
 - Python 3.14+
 - Required Python packages:
 - streamlit
 - python-dotenv
 - google-generativeai
 - PyPDF2
 - python-docx
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